DISTRIBUTIONALLY ROBUST AND DETAIL-FREE DEMAND-RESPONSE CONTRACTING

TEAM A1 — GENERAL CONTRACTING THEORY AND NUMERICAL METHODS

Abstract. The emission-reduction condition depends on the residual-demand distribution, which a regulator never knows exactly. We turn this apparent weakness into two guarantees. *(R1) A worst-case certificate.* Over the natural shape class of distributions, the *uniform* is the least-favorable distribution for carbon: it minimizes the net emission saving. Hence the single closed-form condition $C_R \le C_R^{\dagger}$—derived for the uniform—certifies emission reduction for the entire infinite-dimensional class, a distributionally robust guarantee. *(R2) Detail-free implementation.* Because the optimal contract's thresholds are pinned at *distribution-free* probabilities, a single quantile-targeting mechanism implements the optimum for every distribution: the regulator needs no knowledge of, or prior over, the demand and renewable laws. Both results are verified numerically.

We continue the companion notes `role1_carbon_theory.tex` and `role1_contract_theory.tex`; notation and Assumption 1 are inherited. Two facts proved there drive everything here: the *exact* net-fossil saving

$$\Delta(F) := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = \tfrac{C_M}{R-c}\big(K_F^* - K_M^* - \mu_{\text{low}}^*\big) - \Big(K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ\Big), \quad (1)$$

and the *distribution-free* cutoff probabilities $F(K_F^*; K_R^*) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}$, $F(K_F^* - K_M^*; K_R^*) = \frac{C_M}{R-c}$, $F(K_F^\circ; K_R^\circ) = \varphi^\circ$, which hold for every F. In (1) the conditional means $\mu_{\text{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, K_F^* - K_M^*)]$ and $\mu_{\text{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)]$ are the only channels through which the *shape* of F enters.

1. A distributionally robust emission certificate

Definition 1 (Shape ambiguity set)**.** Let $\mathcal{U}$ be the set of residual-demand distributions F satisfying Assumption 1 such that the density $f(\cdot; K_R^*)$ is non-increasing on the FCL dispatch zone $(-K_R^*, K_F^* - K_M^*)$ and $f(\cdot; K_R^\circ)$ is non-decreasing on the no-FCL fossil zone $(0, K_F^\circ)$.

Economically (companion Remark 10): $\mathcal{U}$ collects grids whose deep-surplus states are more likely than mild-surplus states, and whose high-consumption hours are more likely than low—the empirically typical shape, with rare stress events. The uniform sits on the boundary of $\mathcal{U}$ (constant density).

Theorem 1 (Least-favorable distribution; robust certificate)**.** *Holding the distribution-free cutoff probabilities fixed,*

$$\min_{F \in \mathcal{U}} \Delta(F) \;=\; \Delta(F_{\text{unif}}),$$

attained by the uniform distribution. Consequently emission reduction holds for every $F \in \mathcal{U}$ *if and only if it holds for the uniform, i.e. iff*

$$C_R \le C_R^{\dagger} = \tfrac{1}{2}\Big[c + \tfrac{(R-C_M-c)^2}{R-c} + \tfrac{(R-c-C_F-C_M)^2}{\beta+p-R}\Big]$$

(companion Proposition on $C_R^{\dagger}$). The single closed-form condition is a worst-case guarantee over the infinite-dimensional class $\mathcal{U}$.

Date: Working results — robustness extension.

Proof. From (1), $\partial\Delta/\partial\mu^*_{\text{low}} = -\frac{C_M}{R-c} < 0$ and $\partial\Delta/\partial\mu^\circ_{\text{mid}} = +(\varphi^\circ - q^\circ) > 0$, the latter positive because $\varphi^\circ = F(K^\circ_F; K^\circ_R) > F(0; K^\circ_R) = q^\circ$ (as $K^\circ_F > 0$ and F is increasing in x), so Δ is minimized by the *largest* feasible μ^*_{low} and the *smallest* feasible μ°_{mid}. A non-increasing density on an interval has conditional mean at most the interval midpoint, with equality iff the density is constant; a non-decreasing density has conditional mean at least the midpoint, with equality iff constant. Hence over $\mathcal{U}$, $\sup \mu^*_{\text{low}}$ and $\inf \mu^\circ_{\text{mid}}$ are *both* the respective midpoints, attained *simultaneously* by the uniform. So the uniform jointly minimizes Δ. The closed-form $C^\dagger_R$ is the uniform threshold; if it certifies $\Delta \geq 0$ at the worst case, $\Delta(F) \geq 0$ for all $F \in \mathcal{U}$. □

Remark 1 (Why this is the right robustness statement). Without a shape restriction the ambiguity is vacuous—an adversary placing all conditional mass at an interval endpoint can force $\Delta < 0$, so no distribution-free certificate can exist (verified: unrestricted worst case is negative). $\mathcal{U}$ is the minimal economically meaningful class that restores a certificate, and within it the certificate is *exactly* the uniform closed form. Theorem 1 thus identifies the uniform benchmark of the original paper not as a convenient special case but as the *binding* worst case for a whole family—the strongest reading the closed form can carry.

Remark 2 (Stochastic-order reading). Because Δ moves with the two conditional means, and conditional means on a fixed interval are ordered by the dispersive/likelihood-ratio order, Δ is monotone along any one-parameter tilt of the density toward or away from the uniform. The uniform is the fixed point of this monotone family—the least disperse member consistent with $\mathcal{U}$'s monotonicity.

2. Detail-free implementation

A regulator does not know F. Remarkably, it need not.

Theorem 2 (Prior-independent optimal mechanism)**.** *Define the* quantile-targeting mechanism*: choose capacities so that, at the realized residual-demand law,*

$$\Pr(\tilde{x} \leq K_F) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \quad \Pr(\tilde{x} \leq K_F - K_M) = \frac{C_M}{R - c}, \quad \Pr(\tilde{x} \leq K^\circ_F) = \varphi^\circ,$$

and dispatch $d^*(x) = \min\{(K_F - x)_+, K_M\}$. *For* every *distribution* F *satisfying Assumption 1, this mechanism implements the optimal contract and the optimal capacities. It uses no knowledge of, and no prior over,* F*: only the right-hand-side probabilities, which are functions of the primitives* $(R, c, C_F, C_M, \beta, p)$ *alone.*

Proof. The capacity first-order conditions (companion Lemma) equate F at each threshold to the stated probability; these probabilities contain no distributional object. Targeting them therefore selects the optimal (K^*_R, K^*_F, K^*_M) whatever F is, and Theorem 1 of the contract note makes d^* optimal given those capacities. A regulator can hit the targets from observed empirical quantiles of residual demand without estimating F. □

Remark 3 (Wilson doctrine). Theorem 2 is a detail-free implementation in the sense of the Wilson doctrine: the optimal mechanism is *prior-independent*. The economic content of the distribution-free cutoffs (companion Lemma) is precisely this—the margins $R - c$, $\beta + p - R$, and the no-FCL cutoff probability φ° summarize everything the regulator must know. Distributional detail matters for the *magnitude* of the carbon saving (Theorem 1), never for *how to act*.

Corollary 1 (Robust+detail-free package)**.** *Combining Theorems 1–2: the regulator implements the utility's* profit-maximizing *demand-response contract knowing only six scalar primitives, and—provided the grid lies in the shape class* $\mathcal{U}$ *and* $C_R \leq C^\dagger_R$*—this privately optimal contract is* guaranteed *to lower emissions regardless of the true demand and renewable laws. Emission reduction is a consequence of the profit-optimal contract under the stated condition, not a separate regulatory objective.*

3. Numerical validation

`verify_robust.py`:

- Over 3000 random non-increasing densities on the FCL zone, the conditional mean is $\leq$ midpoint in 3000/3000 cases, with the supremum equal to the midpoint (uniform extremal); symmetrically, over 3000 random non-decreasing densities the conditional mean is $\geq$ midpoint in 3000/3000, infimum at the midpoint.
- Tilting the density continuously from uniform (shape 0) into $\mathcal{U}$, the net carbon saving Δ is *minimized at the uniform*—confirming the uniform is least-favorable (Theorem 1).
- The cutoff probabilities of Theorem 2 are the distribution-free values verified to 10^{-16} on the non-uniform Beta$(2, 2)$ scale-mixture in the companion `verify_exact.py`.

Scope. Strictly A1 (general contracting theory and numerical methods): the object is the utility's profit-maximizing contract under *distributional* ambiguity (ambiguity over the demand/renewable law F), and its prior-independent implementation. This is distinct from—and does not enter—A2 (empirical calibration / magnitudes), B1 (reliability, welfare, and the social-vs-private optimum, including corrective pricing), or B2 (the flexibility continuum, storage, and the multi-grid / price-exogeneity / dynamics robustness-and-limitations discussion). The robustness here is robustness of the *contract* to the unknown distribution, not B2's robustness of the model to market structure.